

```

    "certs":[
      {
        "name":"Fortinet_CA_SSL",
        "cert":"-----BEGIN CERTIFICATE-----\\nMIID5jCCAs6g...Sfu+Q8zE8Crmt6L1X\\bv+q\\n-----END
CERTIFICATE-----\\n"
      },
      {
        "name":"Fortinet_CA_Untrusted",
        "cert":"-----BEGIN CERTIFICATE-----\\nMIID8DCCAtig...3zBbfzP+nVUPC\\nZDPRZA==\\n-----END
CERTIFICATE-----"
      }
    ]
  }
}
"""

```

To verify the secondary FortiGate:

```

FGT_EC_Secondary(global) # diagnose endpoint fctems json deep-inspect-cert-sync
JSON:
"""
{
  "fortigates":[
    "FG2K5E39169*****",
    "FG2K5E39169*****"
  ],
  "vdoms":[
    {
      "vdom":"root",
      "certs":[
        {
          "name":"Fortinet_CA_SSL",
          "cert":"-----BEGIN CERTIFICATE-----\\nMIID5jCCAs6g...Sfu+Q8zE8Crmt6L1X\\bv+q\\n-----END
CERTIFICATE-----\\n"
        },
        {
          "name":"Fortinet_CA_Untrusted",
          "cert":"-----BEGIN CERTIFICATE-----\\nMIID8DCCAtig...3zBbfzP+nVUPC\\nZDPRZA==\\n-----END
CERTIFICATE-----"
        }
      ]
    },
    {
      "vdom":"vdom1",
      "certs":[
        {
          "name":"Fortinet_CA_SSL",
          "cert":"-----BEGIN CERTIFICATE-----\\nMIID5jCCAs6g...Sfu+Q8zE8Crmt6L1X\\bv+q\\n-----END
CERTIFICATE-----\\n"
        },
        {

```

```

        "name": "Fortinet_CA_Untrusted",
        "cert": "-----BEGIN CERTIFICATE-----\\nMIID8CCAfig...3zBbfzP+nVUpC\\nZDPRZA==\\n-----END
CERTIFICATE-----"
    }
  ]
}
]
}
""

```

Allowing FortiClient EMS connectors to trust EMS server certificate renewals based on the CN field

When a FortiGate establishes a Fabric connection with FortiClient EMS, the FortiGate must trust the CA that signed the server certificate. Upon the approval of the EMS certificate, the FortiGate saves the CN field and will trust future certificates that are signed by the same CA and have the same CN field. This allows EMS servers to update their certificates at regular intervals without requiring re-authorization on the FortiGate side, as long as the CN field matches. This prevents interruptions to the EMS Fabric connection when a certificate is updated. Previously, upon the user's approval of the certificate, the certificate fingerprint was saved on the FortiGate, and this required the FortiGate to re-authorize the EMS connection each time the server certificate is updated.

```

config endpoint-control fctems
  edit <id>
    set trust-ca-cn {enable | disable}
  next
end

```

This feature is supported for EMS on-premise and cloud connections, and is the default setting. To authorize based on the certificate fingerprint, disable the `trust-ca-cn` setting. If the setting is changed back to be enabled at a later time, the user will have to re-approve the EMS certificate.

To configure the EMS Fabric connector to trust EMS server certificate renewals based on the CN field:

```

config endpoint-control fctems
  edit 1
    set status enable
    set name "ems133"
    set dirty-reason none
    set fortinetone-cloud-authentication disable
    set server "172.18.62.35"
    set https-port 443
    set serial-number "FCTEMS8822000000"
    set tenant-id "00000000000000000000000000000000"
    set source-ip 0.0.0.0
    set pull-sysinfo enable
    set pull-vulnerabilities enable
    set pull-tags enable
    set pull-malware-hash enable
    set capabilities fabric-auth silent-approval websocket websocket-malware push-ca-certs
  
```

```

common-tags-api tenant-id
    set call-timeout 30
    set out-of-sync-threshold 180
    set websocket-override disable
    set interface-select-method auto
    set trust-ca-cn enable
next
end

```

To verify the configuration:

1. Download the FortiGate configuration file.
2. Verify the ca-cn-info entry, which lists the trusted CA certificate information. In this example, ems133 connector has trust-ca-cn enabled and ems138 connector has trust-ca-cn disabled. For ems138, the ca-cn-info entry does not appear, and there is a certificate-fingerprint field instead:

```

config endpoint-control fcems
    edit 1
        set status enable
        set name "ems133"
        set server "172.18.62.35"
        set serial-number "FCTEMS8822000000"
        set tenant-id "00000000000000000000000000000000"
        set capabilities fabric-auth silent-approval websocket websocket-malware push-ca-certs
common-tags-api tenant-id
    set ca-cn-info "C = CA, ST = BC, L = VANCOUVER, O = FTNT, OU = ReleaseQA, CN =
Release_QA, emailAddress = *****@fortinet.comRelease_QA"
    next
    edit 2
        set status enable
        set name "ems138"
        set server "172.18.62.18"
        set serial-number "FCTEMS8821000000"
        set tenant-id "00000000000000000000000000000000"
        set capabilities fabric-auth silent-approval websocket websocket-malware push-ca-certs
common-tags-api tenant-id
    set certificate-fingerprint
"18:51:76:67:EB:4C:31:A1:51:3F:74:F7:8E:1D:47:5C:18:0F:FE:45:DF:52:91:52:37:0B:27:E7:F1:85:5B:
01:8C:7D:FB:2D:C7:D2:CC:FE:4A:E3:0E:A9:2A:1C:27:4D:D2:A6:C5:87:B8:97:98:57:75:10:15:28:EF:A2:2
3:7C"
    set trust-ca-cn disable
    next
    ...
end

```

3. Run diagnostics to view the certificate information:

```

# diagnose test application fcnacd 96
ems_id 1, certificate authority and common name: C = CA, ST = BC, L = VANCOUVER, O = FTNT, OU
= ReleaseQA, CN = Release_QA, emailAddress = *****@fortinet.comRelease_QA
ems_id 1, fingerprint_sha512:

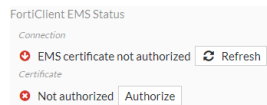
```

```
ems_id 2, certificate authority and common name:
ems_id 2, fingerprint_sha512:
18:51:76:67:EB:4C:31:A1:51:3F:74:F7:8E:1D:47:5C:18:0F:FE:45:DF:52:91:52:37:0B:27:E7:F1:85:5B:0
1:8C:7D:FB:2D:C7:D2:CC:FE:4A:E3:0E:A9:2A:1C:27:4D:D2:A6:C5:87:B8:97:98:57:75:10:15:28:EF:A2:23
:7C
```

FortiClient troubleshooting

Certificate not trusted

When configuring a new connection to an EMS server, the certificate might not be trusted.



When you click *Authorize*, a warning displays: *The server certificate cannot be authenticated with installed CA certificates. Please install its CA certificates on this FortiGate.*

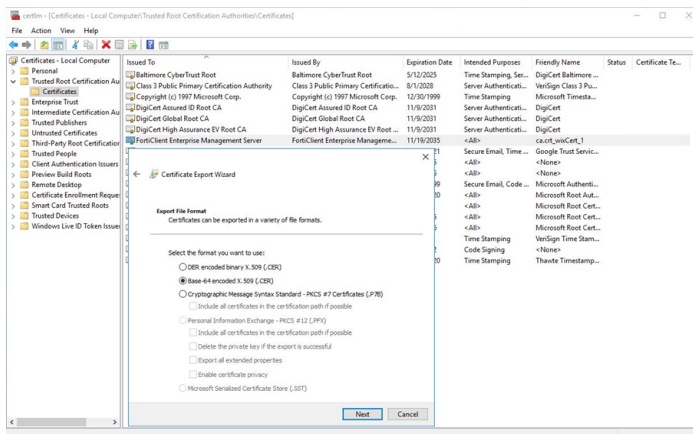
In the CLI, an error message displays when you try to verify the certificate:

```
# execute fctems verify Win2K16-EMS
certificate not configured/verified: 2
Could not verify server certificate based on current certificate authorities.
Error 1--92-60-0 in get SN call: EMS Certificate is not signed by a known CA.
```

The default FortiClient EMS certificate that is used for the SDN connection is signed by the CA certificate that is saved on the Windows server when FortiClient EMS is first installed. You can manually export and install it on the FortiGate.

To manually export and install the certificate on to the FortiGate:

1. Export the EMS certificate on the server that EMS is installed on:
 - a. On the Windows server that EMS is installed on, go to *Settings > Manage computer certificates*.
 - b. In the certificate management module, go to *Trusted Root Certification Authorities > Certificates*.
 - c. Right click on the certificate issued by FortiClient Enterprise Management Server and select *All Tasks > Export*.
 - d. The *Certificate Export Wizard* opens. Click *Next*.
 - e. Select *Base-64 encoded X.509*, then click *Next*.



- f. Enter a file name for the certificate and click *Browse* to select the folder where it will be located, then click *Next*.
- g. Review the settings, then click *Finish*. The certificate is downloaded to the specified folder.
2. On the FortiGate, import the certificate:
 - a. Go to *System > Certificate*. By default, the *Certificate* option is not visible, see [Feature visibility on page 4142](#) for information.
 - b. Click *Import > CA Certificate*.
 - c. Set *Type* to *File*, and click *Upload* to import the certificate from the management computer.
 - d. Click *OK*. The imported certificate is shown in the *Remote CA Certificate* section of the certificate table.
3. Try to authorize the certificate on the FortiGate:
 - a. Go to *Security Fabric > Fabric Connectors* and edit the FortiClient EMS connector. The connection status should now say that the certificate is not authorized.
 - b. Click *Authorize*. The following warning is shown:



The warning can also be seen in the CLI:

```
# execute fctems verify Win2K16-EMS
failure in certificate configuration/verification: -4
Could not verify EMS. Error 1--94-0-401 in get SN call: Authentication denied.
```

4. Authorize the FortiGate on EMS:
 - a. Log in to the EMS server console and go to *Administration > Fabric Devices*.
 - b. Select the serial number of the FortiGate device, then click *Authorize*.
5. Try to authorize the certificate on the FortiGate again:
 - a. On the FortiGate, go to *Security Fabric > Fabric Connectors* and edit the *FortiClient EMS* card.
 - b. Click *Refresh*.
 - c. When presented with the EMS server certificate, click *Accept* to accept the certificate. Your connection should now be successful and authorized.
 - d. Click *OK*.

Synchronizing FortiClient ZTNA tags

ZTNA tags (formerly FortiClient EMS tags in FortiOS 6.4 and earlier) are tags synchronized from FortiClient EMS as dynamic address objects on the FortiGate. FortiClient EMS uses zero-trust tagging rules to automatically tag managed endpoints based on various attributes detected by the FortiClient. When the FortiGate establishes a connection with the FortiClient EMS server through the EMS Fabric connector, it pulls zero-trust tags containing device IP and MAC addresses and converts them to read-only dynamic address objects. It also establishes a persistent WebSocket connection to monitor for changes in zero-trust tags, which keeps the device information current. These ZTNA tags can then be used in ZTNA rules, firewall rules, and NAC policies to perform security posture checks. ZTNA tags are displayed in the *Device Inventory* widget, *FortiClient* widget, and *Asset Identity Center* page.



By enabling ZTNA EMS tag checking in a firewall policy, you can include EMS tag information in the traffic log.

When primary and secondary ZTNA EMS tag checking is enabled using address groups, the *Primary EMS tag* and *Secondary EMS tag* fields will be included in the GUI traffic logs. Likewise, the *emstag* and *emstag2* fields will be included in the CLI traffic logs.

When using WebSocket, EMS pushes notifications to the corresponding FortiGate when there are updates to tags or other monitored attributes. The FortiGate then fetches the updated information using the REST API over TCP/8013. When WebSocket is not used (due to an override or unsupported EMS version), updates are triggered on demand from the FortiGate side over the REST API.

If the WebSocket capability is detected, the capabilities setting will automatically display the WebSocket option. You can use the `diagnose test application fcnacd 2` command to view the status of the WebSocket connection.

In the following example, the FortiGate connects to and retrieves ZTNA tags from a FortiClient EMS configured with tagging rules. It is assumed that zero-trust tags and rules are already created on the FortiClient EMS. For more information, see the [Zero Trust Tags](#) section of the EMS Administration Guide.

To verify zero-trust tags in FortiClient EMS:

1. Go to *Zero Trust Tags > Zero Trust Tagging Rules* to view the tags.

FortiClient Endpoint Management Server

Invitations 0 11 admin

Dashboard

Endpoints

Deployment & Installers

Endpoint Policy & Components

Endpoint Profiles

Zero Trust Tags

Quarantine Management

Administration

System Settings

Zero Trust Tagging Rules

Zero Trust Tag Monitor

Fabric Device Monitor

Zero Trust Tagging Rule Sets

Import Signatures

Manage Tags

+ Add

Refresh

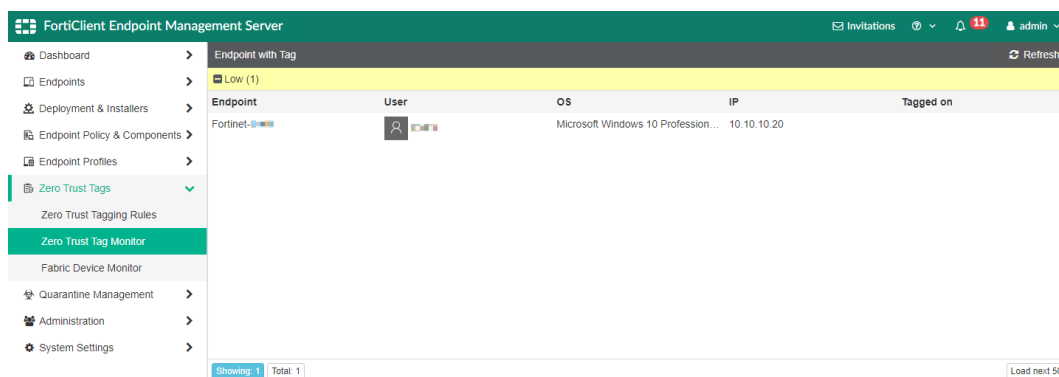
Clear Filters

Name	Tag	Enabled	Comments
Malicious-File-Detected	Malicious-File-Detected		Detect presence of a malicious file

Showing: 1

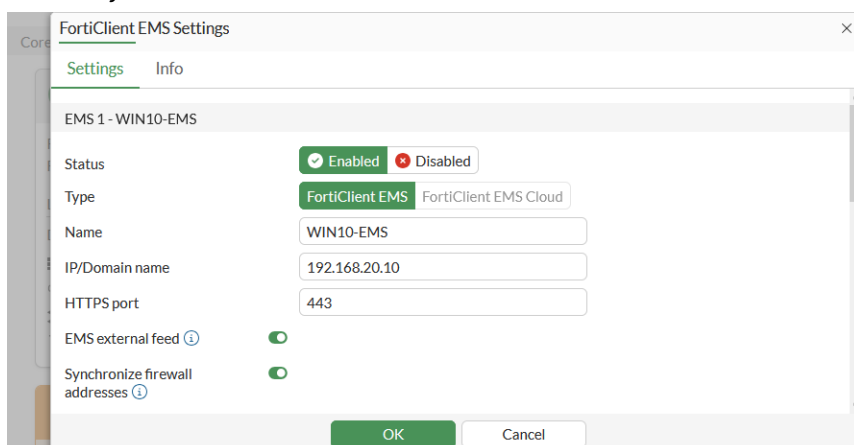
Total: 1

2. Go to *Zero Trust Tags > Zero Trust Tag Monitor* to view the registered users who match the defined tag.



To configure the EMS Fabric connector to synchronize ZTNA tags in the GUI:

1. Configure the EMS Fabric connector:
 - a. On the root FortiGate, go to *Security Fabric > Fabric Connectors* and double-click the *FortiClient EMS* card.
 - b. Set the *Status* to *Enabled* for one of the EMS servers.
 - c. Enable *Synchronize firewall addresses*.



- d. Configure the other settings as needed and validate the certificate.
 - e. Click *OK*.
2. Enable ZTNA:
 - a. Go to *System > Feature Visibility* and enable *Zero Trust Network Access*.
 - b. Click *Apply*.
3. Go to *Policy & Objects > ZTNA* and select the *ZTNA Tags* tab. You will see the ZTNA IP and ZTNA MAC tags synchronized from the FortiClient EMS.

ZTNA Rules ZTNA Servers ZTNA Tags			
+ Create New Group Edit Delete Search			
Name	Details	Comments	Ref.
ZTNA IP Tag			
Zero-day Detections			0
Medium			0
Malicious-File-Detected			2
Low			4
IOC Suspicious			0
High			0
FCITEMS_ALL_FORTICLOUD_SERVERS			0
Critical			1
all_registered_clients			1
ZTNA MAC Tag			
Zero-day Detections			0
Medium			0
Malicious-File-Detected			0
Low			0
IOC Suspicious			0
High			0
Critical			0
all_registered_clients			0
ZTNA Tag Group			
grp_ems138	all_registered_clients Critical		0

To configure the EMS Fabric connector to synchronize ZTNA tags in the CLI:

1. Configure the EMS Fabric connector on the root FortiGate:

```
config endpoint-control fcitems
  edit 1
    set status enable
    set name "WIN10-EMS"
    set server "192.168.20.10"
    set https-port 443
    set pull-sysinfo enable
    set pull-vulnerabilities enable
    set pull-tags enable
    set pull-malware-hash enable
    set capabilities fabric-auth silent-approval websocket
  next
end
```

2. Verify which IPs the dynamic firewall address resolves to:

```
# diagnose firewall dynamic list
List all dynamic addresses:
FCITEMS0000100000_all_registered_clients: ID(51)
  ADDR(172.17.194.209)
  ADDR(10.10.10.20)
...

FCITEMS0000100000_Low: ID(78)
  ADDR(172.17.194.209)
  ADDR(10.10.10.20)
...

FCITEMS0000100000_Malicious-File-Detected: ID(190)
```


ADDR(172.17.194.209)
ADDR(10.10.10.20)
...



When running the FortiGate in multi-VDOM mode, by default, EMS is configured in the global VDOM. All ZTNA tags synchronized with the globally configured EMS are shared by all VDOMs. FortiOS 7.4 and later supports configuring EMS on a per-VDOM basis. See [Configuring FortiClient EMS and FortiClient EMS Cloud on a per-VDOM basis](#) for more information.

Configuring LAN edge devices

The *LAN Edge Devices* connector card displays a summary about the FortiGates, FortiAPs, FortiSwitches, and FortiExtenders in the Security Fabric. Information about the device type, number of devices, and number of unregistered and unauthorized devices is displayed. If there are devices that do not have a green checkmark in the *Status* column, hover over the status message to view the tooltip with required action. In this example, there are FortiAPs and FortiSwitches that are not registered. The tooltip includes a link to the *System > Firmware & Registration* page to register or authorize the devices. The tooltip for FortiExtenders includes a link to the *Network > FortiExtenders > Managed FortiExtenders* page to register or authorize the devices.

Core Network Security Connectors

Security Fabric Setup

Logging & Analytics

FortiClient EMS

Role	Fabric Root
Fabric name	fabric

FortiAnalyzer	Cloud Logging	Status
Enabled	Enabled	Enabled
Disabled	Disabled	Disabled

EMS	IP address	Status
10.100.88.5	10.100.88.5	Enabled

LAN Edge Devices

Device Type	Device Count	Status
FortiGate	5	All authorized & registered
FortiAP	3	3 devices not registered
FortiSwitch	3	3 devices not registered
FortiExtender	0	None configured

Go to the *Firmware & Registration* page to authorize your FortiGate and handle any other device related concerns.

[Firmware & Registration](#)



If the default auto-auth-extension-device settings on the FortiAP or FortiSwitch have been modified, manual authorization in the Security Fabric may not be required.

For more information about configuring FortiAPs, see [Configuring the FortiGate interface to manage FortiAP units](#) and [Discovering, authorizing, and deauthorizing FortiAP units](#).

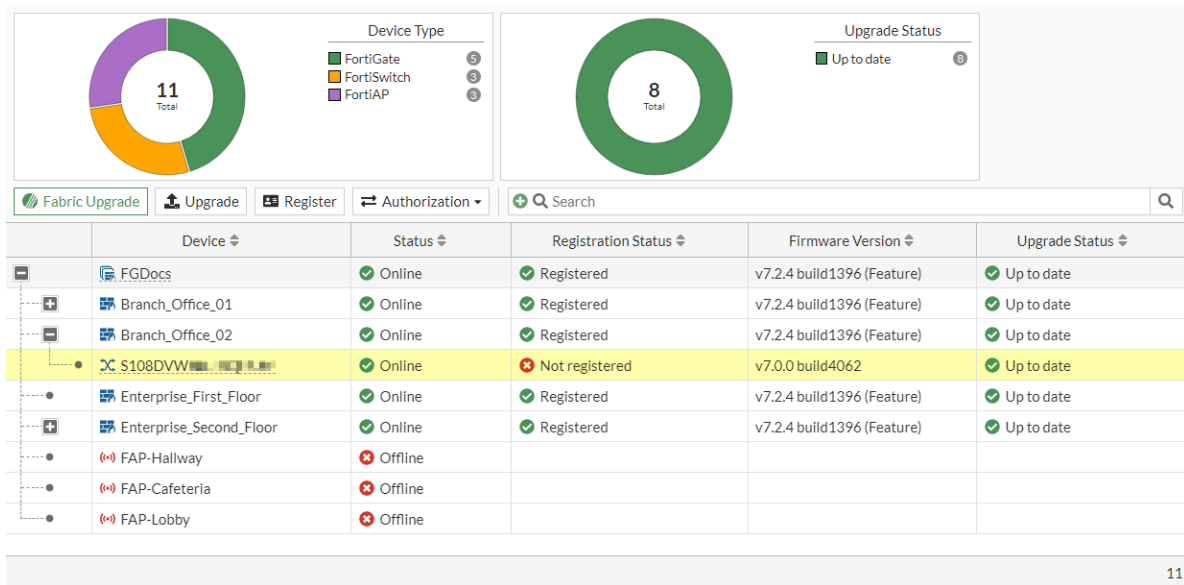
For more information about configuring FortiSwitches, see [Discovering, authorizing, and deauthorizing FortiSwitch units](#).

For more information about configuring FortiExtenders, see [Adding a FortiExtender on page 791](#) and [FortiExtender and FortiGate integration](#).

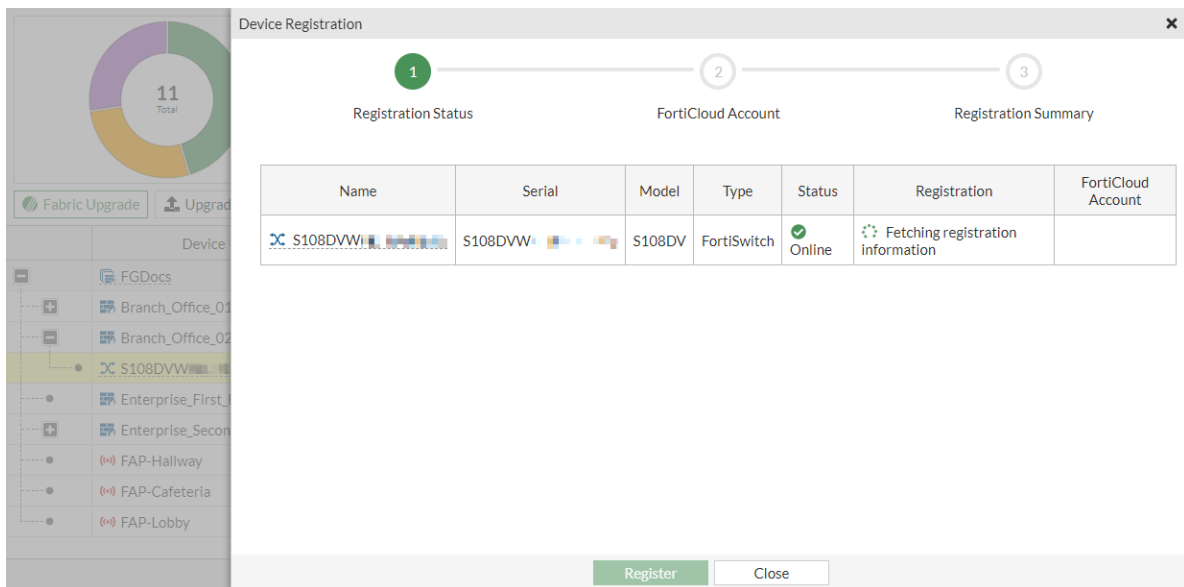
The following example shows how to register a FortiSwitch that has been authorized to join the Security Fabric. The procedure is similar for FortiAPs and FortiExtenders. If the device requires authorization, refer to [Authorizing devices on page 3960](#).

To register a FortiSwitch:

1. Connect the device to the root FortiGate.
2. On the root FortiGate, go to *System > Firmware & Registration*. (Use the same page for a FortiAP, or *Network > FortiExtenders > Managed FortiExtenders* for a FortiExtender.)
3. Select the unregistered device and click *Register*.



The *Device Registration* pane opens and the FortiGate verifies the registration information.



4. Once the registration process is complete, click *Close*.

Configuring central management

The *Central Management* Fabric connector card on the root FortiGate is used to configure the FortiManager settings, which includes on-premises FortiManager, FortiGate Cloud, and FortiManager Cloud. After the *Central Management* connector is configured, it automatically synchronizes with any connected downstream devices.

This topic covers the following central management aspects:

- [Configuring FortiManager](#)
- [Configuring FortiManager Cloud](#)

Configuring FortiManager

Once the *Central Management* Fabric connector is configured, the root FortiGate pushes this configuration to downstream FortiGates. FortiManager provides remote management of FortiGate devices over TCP port 541. The FortiManager must have internet access for it to join the Security Fabric.

Once configured, the FortiGate can receive antivirus and IPS updates, and allows remote management through FortiManager or the FortiGate Cloud service. The FortiGate management option must be enabled so that the FortiGate can accept management updates to its firmware and FortiGuard services.

Adding a FortiManager device to the Security Fabric requires the following steps in FortiOS, which can be completed in the GUI or CLI:

- Specify the FortiManager IP address or domain name.
- Approve the FortiManager serial number returned by the FortiManager server certificate. This ensures that the administrator is connecting the FortiGate to the desired FortiManager.

After completing the steps in FortiOS, go to FortiManager to complete the process by authorizing the FortiGate.

To enable management of FortiGate VM devices in FortiManager:

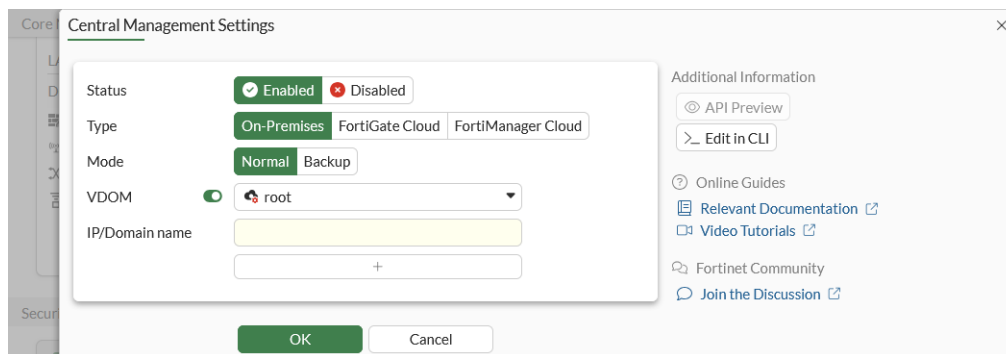
If you are using a FortiGate VM, enable recognition of FortiGate VM serial numbers in FortiManager. For FortiGate hardware models, skip this step. See [Enabling management of VM devices](#) in the FortiManager Administration Guide for more information.

In FortiManager, run the following command:

```
config sys global
    set fgfm-allow-vm enable
end
```

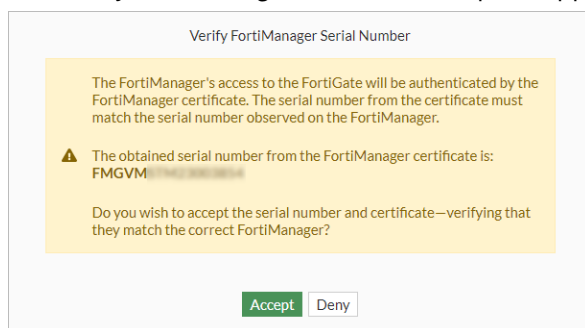
To add a FortiManager to the Security Fabric in the GUI:

1. On the root FortiGate, go to *Security Fabric > Fabric Connectors* and double-click the *Central Management* card.
2. Set the *Status* to *Enabled*.
3. Set the *Type* to *On-Premises*.



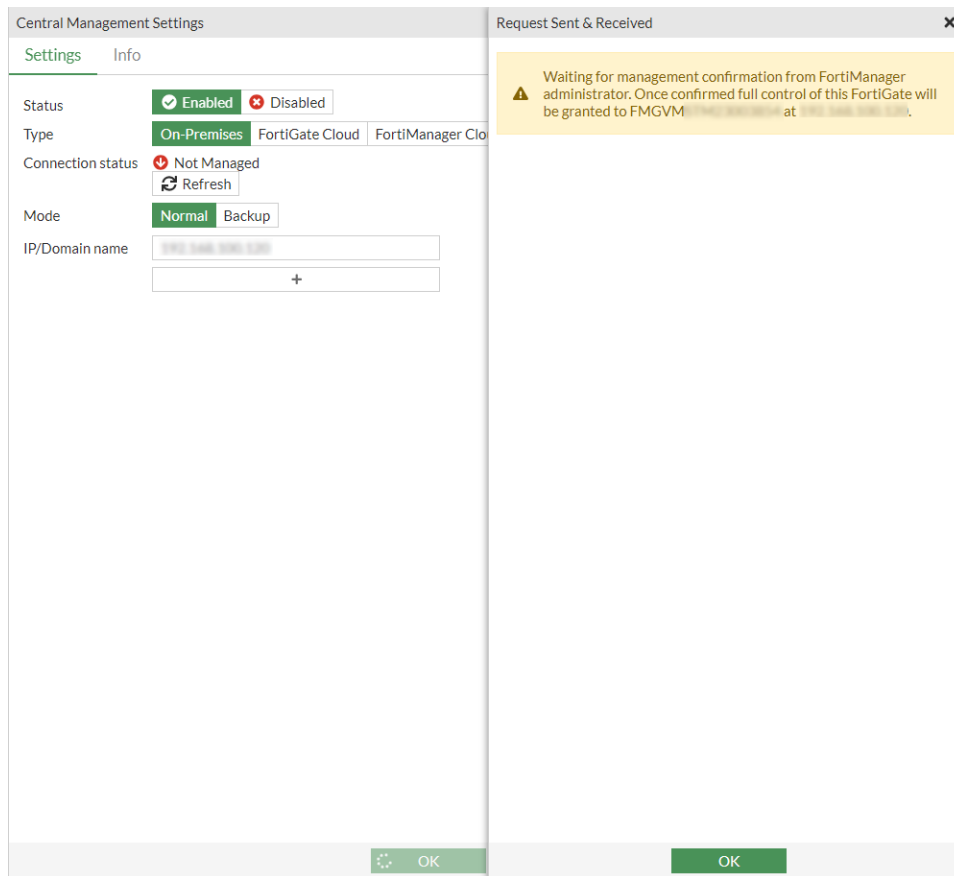
4. If VDOMs are enabled, enable VDOM and select the VDOM that has a route to the FortiManager.
5. Enter the *IP/Domain Name* of the FortiManager.
6. Click *OK*.

The *Verify FortiManager Serial Number* pane appears.

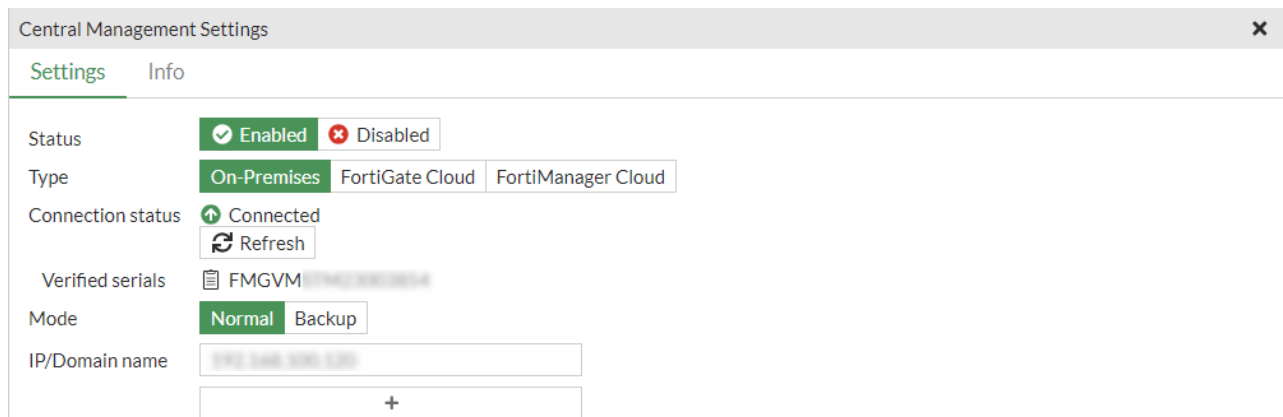


7. Review the serial number, and click *Accept*.

The *Request Sent & Received* pane appears, indicating the FortiGate must be authorized on FortiManager.



8. Go to FortiManager and authorize the FortiGate. See [Authorizing the FortiGate in FortiManager on page 4573](#).
9. After the FortiGate is registered, log in to FortiGate again as either read-only or read/write.
10. Go to *Security Fabric > Fabric Connectors* and double-click the *Central Management* card. The *Connection Status* is updated to *Connected*.



To add a FortiManager to the Security Fabric in the CLI:

1. Enter the FortiManager connection information:

```
config system central-management
  set type fortimanager
  set fmg {<IP_address> | <Domain name>}
end
```

The Serial Number for FortiManager is not entered.

In order to verify identity of FortiManager serial number is needed.

If serial number is not set, connection will be set as unverified.

FortiGate can establish a connection to obtain the serial number now. Do you want to try to connect now? (y/n)y

Obtained serial number from FortiManager 172.16.200.1 is: FMGVMSTM2300xxxx

Do you confirm that this is the correct serial number? (y/n)y

Successfully registered to FortiManager. This device may need to be authorized on FortiManager.

Auto firmware upgrade in system.fortiguard has been paused since this FortiGate is now managed by FortiManager. The upgrade will resume automatically when this FortiGate is released from FortiManager. The upgrade status may be viewed using the following command `diagnose test application forticldd 13`.

Any pending automatic patch-level firmware upgrade has been removed

2. Go to FortiManager and authorize the FortiGate. See [Authorizing the FortiGate in FortiManager on page 4573](#).
3. If necessary on FortiGate, use the `diagnose fdsm central-mgmt-status` command to diagnose the connection.

- If the connection is not yet successful because the FortiManager serial number is not verified, the following information is displayed:

```
# diagnose fdsm central-mgmt-status
Connection status: Handshake
Registration status: Unknown
Serial: FMGVMSTM2300xxxx
```

- If the connection is up, but the FortiGate has not been authorized by FortiManager, the following information is displayed:

```
# diagnose fdsm central-mgmt-status
Connection status: Up
Registration status: Unregistered
Serial: FMGVMSTM2300xxxx
```

- If the connection is up, and the FortiGate has been authorized, the following information is displayed:

```
# diagnose fdsm central-mgmt-status
Connection status: Up
Registration status: Registered
Serial: FMGVMSTM2300xxxx
```

Authorizing the FortiGate in FortiManager

After completing the GUI or CLI steps in FortiOS, go to FortiManager to authorize the FortiGate to complete the process.

To authorize the FortiGate in FortiManager:

1. On FortiManager, go to *Device Manager* and find the FortiGate in the *Unauthorized Devices* list.
The unauthorized device list is located in the root ADOM, regardless of the firmware version of the root ADOM or FortiOS.
2. Select the FortiGate device or devices, and click *Authorize* in the toolbar.
3. In the *Authorize Device* pop-up, adjust the device names as needed, select the appropriate ADOM (if applicable), and click *OK*.

For more information about using FortiManager, see the [FortiManager Administration Guide](#).

Configuring FortiManager Cloud

This cloud-based SaaS management service is available through FortiManager. This service is included in FortiCloud accounts with a FortiManager Cloud account level subscription (ALCI).

Configuring a per-device license

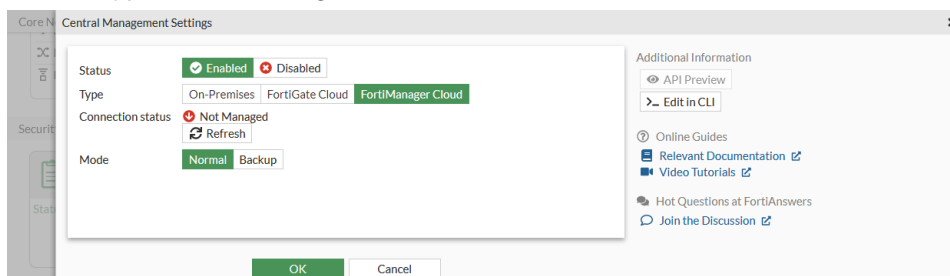
Once the FortiGate has acquired a contract named *FortiManager Cloud*, FortiCloud creates a cloud-based FortiManager instance under the user account. You can launch the portal for the cloud-based FortiManager from FortiCloud, and its URL starts with the User ID.

You can use a FortiGate with a contract for *FortiManager Cloud* to configure central management by using the FQDN of *fortimanager.forticloud.com*. A FortiGate-FortiManager tunnel is established between FortiGate and the FortiManager instance.

After the tunnel is established, you can execute FortiManager functions from the cloud-based FortiManager portal.

To configure FortiManager Cloud central management:

1. Enable FortiManager Cloud.
 - a. Go to *Security Fabric > Fabric Connectors* and double-click the *Central Management* card.
 - b. Set the *Status* to *Enabled*.
 - c. Set the *Type* to *FortiManager Cloud*.



- d. Click *OK*.



The *FortiManager Cloud* button can only be selected if you have a FortiManager Cloud product entitlement.

2. In the FortiManager Cloud instance, go to *Device Manager* and authorize the FortiGate. See [Authorizing devices](#) for more information.

When using the FortiGate to enable FortiManager Cloud, the FortiGate appears as an unauthorized device. After authorizing the FortiGate, it becomes a managed device.

In FortiOS, the *Security Fabric > Fabric Connectors* page now displays green arrow in the *Central Management* card because FortiManager Cloud is registered.

Diagnostics

To verify the contract information:

```
# diagnose test update info contract
...
System contracts:
...
Account contracts:
    FMGC,Thu Dec  2 16:00:00 2022
...
```

To verify the FortiManager Cloud instance has launched and the FortiGate is registered:

```
# diagnose fdsm central-mgmt-status
Connection status: Up
Registration status: Registered
Serial: FMGVMSTM2300xxxx
```

Configuring sandboxing

The Security Fabric supports the following types of FortiSandbox deployments:

Type	Description	Requirements	Next steps
FortiGate Cloud Sandbox (FortiSandbox SaaS)	Files are sent to Fortinet's Cloud Sandbox cluster for post-processing.	The FortiGate must be subscribed to the Advanced Malware Protection (AMP) license, which includes the AV license.	Activate and configure your FortiGate Cloud Sandbox. Use post-transfer scan with Antivirus.

Type	Description	Requirements	Next steps
FortiGuard Inline Malware Prevention System	Files are sent to Fortinet's Cloud Sandbox cluster for real-time processing.	The FortiGate must either be subscribed to the Enterprise Protection bundle or have an a la carte Inline Malware Prevention license.	Verify your license. - Configure your FortiGate. Use Inline Scan with Antivirus. Verify Sandbox detection.
FortiSandbox Cloud (FortiSandbox PaaS)	Files are sent to a dedicated FortiCloud hosted instance of FortiSandbox for processing.	- FortiCloud premium license. - FortiSandbox Cloud entitlement. - The FortiGate and FortiCloud licenses are registered to the same account.	Provision your dedicated instance. - Configure your FortiGate for post-transfer scan or inline scan. - Verify Sandbox detection.
FortiSandbox Appliance	Files are sent to a physical or VM appliance, typically residing on premise, for processing.	None	Configure your FortiSandbox. - Configure your FortiGate for post-transfer scan or inline scan. - Verify Sandbox detection.

To apply sandboxing in a Security Fabric, connect one of the FortiSandbox deployments, then configure an antivirus profile to submit files for dynamic analysis. The submission results supplement the AV signatures on the FortiGate. FortiSandbox inspection can also be used in web filter profiles.

In a Security Fabric environment, sandbox settings are configured on the root FortiGate. Once configured, the root FortiGate pushes the settings to other FortiGates in the Security Fabric.

FortiGate Cloud Sandbox (FortiSandbox SaaS)

FortiGate Cloud Sandbox allows users to take advantage of FortiSandbox features without having to purchase, operate, and maintain a physical appliance. It also allows you to control the region where your traffic is sent to for analysis. This allows you to meet your country's compliance needs regarding data storage locations.

Users are not required to have a FortiCloud account to use FortiGate Sandbox Cloud.

The submission to the cloud with a valid FortiGuard Antivirus (AVDB) license is rate limited per FortiGate model. Refer to the Service Description for details. For those without an AVDB license, the submission is limited to only 100 per day.

To configure FortiGate Cloud Sandbox, you must first activate the connection. Note that FortiGate Cloud Sandbox is decoupled from FortiGate Cloud logging, so you do not need to have a FortiCloud account or have cloud logging enabled.



To ensure proper connectivity to FortiGate Cloud Sandbox, on the FortiGate in *Security Profiles > AntiVirus*, create a profile with *Send files to FortiSandbox for inspection* configured, and create a firewall policy with logging enabled that uses the Sandbox-enabled AV profile.

To enable FortiGate Cloud Sandbox in the GUI:

1. Go to *Security Fabric > Fabric Connectors* and double-click the *Sandbox* card.
2. Set the *Status* to *Enabled*.
3. Set the *Type* to *FortiGate Cloud*.



If the *FortiSandbox Cloud* option is grayed out or not visible, enter the following in the CLI:

```
config system global
    set gui-fortigate-cloud-sandbox enable
end
```

4. Select a *Region* from the dropdown.
5. Click *OK*.

To activate the FortiGate Cloud Sandbox connection:

```
# execute forticloud-sandbox region
0 Europe
1 Global
2 Japan
3 US
Please select cloud sandbox region[0-3]:3
```

After a region is selected, the following configuration is added:

```
config system fortiguard
    set sandbox-region <region>
end
```

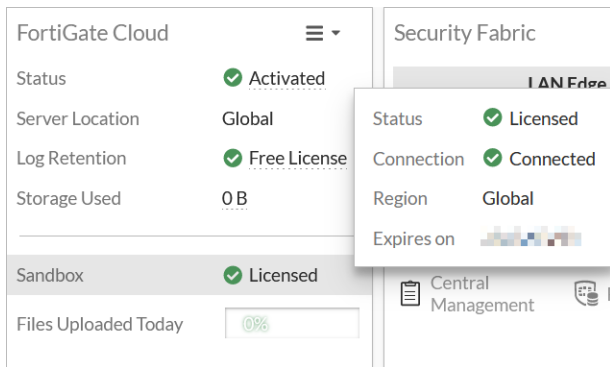


Alternatively, using the `execute forticloud-sandbox update` command also works.

To obtain or renew a FortiGuard antivirus license:

1. See the [How to Purchase or Renew FortiGuard Services](#) video for FortiGuard antivirus license purchase instructions.
2. Once a FortiGuard license is purchased and activated, users are provided with a paid FortiSandbox Cloud license.

- a. Go to *Dashboard > Status* to view the FortiSandbox Cloud license indicator.



- b. Alternatively, go to *System > FortiGuard* to view the FortiSandbox Cloud license indicator.

FortiSandbox Cloud (FortiSandbox PaaS)

FortiSandbox Cloud offers more features and better detection capability. Connecting to FortiSandbox Cloud will automatically use the cloud user ID of the FortiGate to connect to the dedicated FortiSandbox Cloud instance. The FortiGate automatically detects if there is a valid entitlement.

The following items are required to initialize FortiSandbox Cloud:

- A FortiCloud premium account.
- A valid FortiSandbox Cloud contract on the FortiGate. To view contract information in the CLI, enter `diagnose test update info`. The User ID at the end of the output shows FortiCloud which FortiSandbox Cloud account the FortiGate is connected to.
- A provisioned FortiSandbox Cloud. See [Deploying FortiSandbox Cloud](#) for information.

To configure FortiSandbox Cloud in the GUI:

1. Go to *Security Fabric > Fabric Connectors* and double-click the *Sandbox* card.
2. Set the *Status* to *Enabled*.
3. Set the *Type* to *FortiSandbox Cloud*.

 If the *FortiSandbox Cloud* option is grayed out or not visible, enter the following in the CLI:

```
config system global
    set gui-fortigate-cloud-sandbox enable
end
```

4. Click *OK*.

To configure FortiSandbox Cloud in the CLI:

```
config system fortisandbox
    set status enable
    set forticloud enable
```

```
set server fortisandboxcloud.com
end
```

If the FortiGate does not detect the proper entitlement, a warning is displayed and the CLI configuration will not save.

If the FortiSandbox Cloud is running version 4.0.0 and later, the FortiGate will automatically connect to fortisandboxcloud.com, and then discover the specific region and server to connect to based on which region the customer selected to deploy their FortiSandbox Cloud instance. The FortiGate must have a FortiCloud premium account license and a FortiSandbox Cloud VM license for this functionality.

FortiSandbox appliance

FortiSandbox appliance is the on-premise option for a full featured FortiSandbox. Connecting to a FortiSandbox appliance requires that Cloud Sandbox is disabled.

FortiOS can authenticate FortiSandbox using Fortinet CA certificates. Third-party certificates are not supported.

By default, FortiOS uses an automatic certificate verification process to authenticate FortiSandbox. In the fabric connector, enable FortiSandbox and enter its IP address. FortiOS connects to FortiSandbox to retrieve and display its serial number for verification. Once verified, FortiOS adds the serial number to the cn (Common Name) field.

You can also disable automatic certificate verification, and use the CLI to manually configure CA (Certificate Authority) and CN fields for FortiSandbox.

FortiOS uses the following methods with FortiSandbox: post transfer and inline. Certificate checks and verification have only been implemented for the post-transfer method.

To enable FortiSandbox appliance in the GUI:

1. Go to *Security Fabric > Fabric Connectors* and double-click the *Sandbox* card.
2. Set the *Status* to *Enabled*.
3. In the *Server* field, enter the FortiSandbox device's IP address.

Sandbox Settings

Status: Enabled Disabled

Type: FortiSandbox FortiSandbox Cloud

Server:

Connection status: Refresh Authorize

Notifier email:

Inline scan: ☐

Verify FortiSandbox certificate: ☒

Dynamic Malware Detection
Version: not loaded

URL Threat Detection
Version: 0.0

Entries
0 View

Sandbox Statistics (last 7 days)

File type	Detected
Total submitted	0
Critical (Malicious)	0
High Risk	0
Medium Risk	0
Low Risk	0
Clean	0

Additional Information
API Preview

OK Cancel

4. Optionally, enter a *Notifier email*.
5. Click **OK**. A verification message is displayed.

Verify FortiSandbox Serial Number

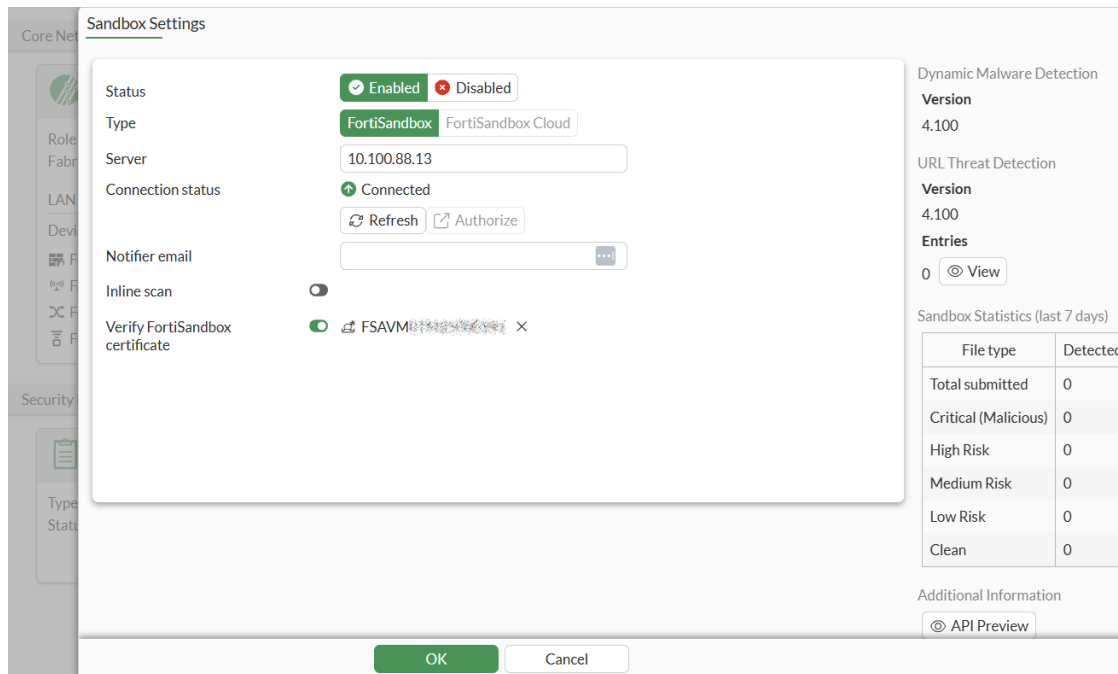
The FortiSandbox's access to the FortiGate will be authenticated by the FortiSandbox certificate. The serial number from the certificate must match the serial number observed on the FortiSandbox.

⚠ The obtained serial number from the FortiSandbox certificate is:
FSAVM...

Do you wish to accept the serial number and certificate—verifying that they match the correct FortiSandbox?

Accept Deny

6. Click **Accept** to verify the correct serial number.
The serial number is displayed beside *Verify FortiSandbox certificate*, and the *Connection status* is *Connected*.



To enable FortiSandbox appliance in the CLI:

1. Enable FortiSandbox and configure its IP address:

```
config system fortisandbox
  set status enable
  set forticloud disable
  set server "172.18.70.76"
end
```

2. After the message is displayed, type y to establish a connection and retrieve the FortiSandbox serial number:

```
In order to verify identity of FortiSandbox serial number is needed.
If serial number is not set, connection will be set as unverified and
access to local config and files will be accessible only with user name/password.
FortiGate can establish a connection to obtain the serial number now. Do you want to try to
connect now? (y/n)y
```

3. After FortiOS obtains the FortiSandbox serial number, type y to verify it:

```
Obtained serial number from X509 certificate of FortiSandbox is: FSA3KET321000049
Serial number from certificate MUST be the same as serial number observed in FortiSandbox.
If these two serial numbers don't match, connection will be dropped.
Please make sure the serial numbers are matching.
Do you confirm that this is the correct serial number? (y/n)y
```

FortiOS adds the serial number to the cn field.

4. Show the FortiSandbox settings to verify the CN field: